



**BALL STATE  
UNIVERSITY**

---

# **Module 4 Lecture - Discrete Random Variables**

Introduction to Statistical Methods

---

Quinton Quagliano, M.S., C.S.P

Department of Educational Psychology

## Table of Contents

|          |                                                                               |          |
|----------|-------------------------------------------------------------------------------|----------|
| <b>1</b> | <b>Overview and Introduction</b>                                              | <b>2</b> |
| 1.1      | Textbook Learning Objectives . . . . .                                        | 2        |
| 1.2      | Instructor Learning Objectives . . . . .                                      | 2        |
| 1.3      | Introduction . . . . .                                                        | 2        |
| <b>2</b> | <b>Probability Distribution Function (PDF) for a Discrete Random Variable</b> | <b>3</b> |
| 2.1      | Introduction . . . . .                                                        | 3        |
| 2.2      | PDF Tables . . . . .                                                          | 4        |
| <b>3</b> | <b>Mean or Expected Value and Standard Deviation</b>                          | <b>5</b> |
| 3.1      | Introduction . . . . .                                                        | 5        |
| 3.2      | Mean Calculation Example . . . . .                                            | 6        |
| 3.3      | Standard Deviation Calculation Example . . . . .                              | 6        |
| 3.4      | Section Conclusion . . . . .                                                  | 7        |
| <b>4</b> | <b>Binomial Distribution</b>                                                  | <b>7</b> |
| 4.1      | Introduction . . . . .                                                        | 7        |
| 4.2      | Notation . . . . .                                                            | 8        |
| <b>5</b> | <b>Conclusion</b>                                                             | <b>8</b> |
| 5.1      | Recap . . . . .                                                               | 8        |
| 5.2      | Lecture Check-in . . . . .                                                    | 8        |

# 1 Overview and Introduction

## 1.1 Textbook Learning Objectives

- Recognize and understand discrete probability distribution functions, in general.
- Calculate and interpret expected values.
- Recognize the binomial probability distribution and apply it appropriately.
- Classify discrete word problems by their distributions.

## 1.2 Instructor Learning Objectives

- Refresh on understanding of behavior and characteristics of discrete variables
- Understand the how to write probability distribution function tables and use them in calculation
- Be able to continue to write coherent and readable notation for probability problems

## 1.3 Introduction

- We will be expanding on what we spoke about with \_\_\_\_\_ in the last lecture, but now explicitly applying those ideas to variables

 Discuss: Try describing discrete variables in your own words, based on what you know from Module 1

- We previously described \_\_\_\_\_ data, or that which is numeric and “count-able”, that has no intervals between integers.
  - Those will be the focus of this lecture, before we focus on \_\_\_\_\_ variables in the next lecture
- In this lecture, we will introduce the concept of **random variables**, or those that can vary in subsequent \_\_\_\_\_ (used in the sense as how it was introduced during the probability lecture)
- The \_\_\_\_\_ of random variables is as follows:
  - Uppercase alphabetical character, e.g.  $X$ ,  $Y$ ,  $Z$ , etc. → written/verbose description of variable

- Lowercase alphabetical equivalent, e.g.  $x$ ,  $y$ ,  $z$ , etc. → the possible values  $X$  can take on
- Example from book:
  - $X$  = the number of heads you get when you toss three \_\_\_\_\_ coins (concrete description of what the variable is)
  - $x = 0, 1, 2, 3$  (what values it could be)
  - This is a discrete, random variable, because  $x$  has \_\_\_\_\_ different values (random) and because those different values follow the discrete rules, i.e. “countable” in the sense that it is a number of times and no intervals between integers

? Review: What is a synonym for 'fair' when discussing probability?

- ☒ A) Equally similar
- ☐ B) Similar
- ☐ C) Equivalent
- ☐ D) Equally likely

Explanation:

## 2 Probability Distribution Function (PDF) for a Discrete Random Variable

### 2.1 Introduction

- A **probability distribution function** or PDF is a very scary way to say the list of possible  $x$  values and their respective \_\_\_\_\_
  - The \_\_\_\_\_ they follow are simple:
    1. All probabilities must sum up to 1, or  $\sum P(x) = 1$
    2. Each probability is between 0 and 1, inclusive, or  $0 \leq P(x) \leq 1$

 Discuss: Review: try writing the mathematical notation for probability of event A equalling a 30% chance of occurring

- In a probability distribution function, we can think of each   value of  $X$ , i.e., the  $x$ s, having some specified probability of occurring

## 2.2 PDF Tables

- Example:
  - A survey is being done on how many vehicles families own. Assume the following probability distribution function:
  - $X$  = the number of vehicles owned by a family
  - $x = 0, 1, 2, 3, 4$

| $x$ | $P(X = x)$ |
|-----|------------|
| 0   | 0.10       |
| 1   | 0.20       |
| 2   | 0.30       |
| 3   | 0.25       |
| 4   | 0.15       |

or...

| $x$ | $P(x)$              |
|-----|---------------------|
| 0   | $P(x = 0) = 10/100$ |
| 1   | $P(x = 1) = 20/100$ |
| 2   | $P(x = 2) = 30/100$ |
| 3   | $P(x = 3) = 25/100$ |
| 4   | $P(x = 4) = 15/100$ |

- In this example, If I were to randomly select a single family from this distribution, I would have a 30% chance of picking a family with 2 cars

 Discuss: On 30% of days I bring one umbrella with me, and on 70% of days I bring no umbrella with me. Write out a PDF table as above to demonstrate this probability distribution function

- There are 4 types of distributions for discrete random variables:
  - Distributions
  - Distributions
  - Distributions
  - Distributions
  - Each of these has their place, but the last 3 come into play more on more  statistical techniques - we'll focus just on the binomial distribution for this lecture

**! Important**

Just because I leave these distribution out in this lecture, doesn't mean that they aren't important! It's mostly just that the others won't be as useful immediately to beginners in statistics.

### 3 Mean or Expected Value and Standard Deviation

### 3.1 Introduction

- With PDFs, we sometimes may wish to find the **expected value**, or the “**long-term**” **average or mean**. Thus, doing running this experiment over and over again, we’d expect to                      on this expected mean
- This is based in the **Law of Large Numbers** a topic alluded to several times in the previous modules
  - Somewhat review from the                      module: this law states that relative observed frequency approaches the theoretical probability as the number of experiments or trials increases
- For a discrete probability function:

$$\mu = \sum (x \cdot P(x))$$

$$\sigma = \sqrt{\sum [(x - \mu)^2 \cdot P(x)]}$$

- If outcomes for the experiment are equally likely then these formulas work to find the mean and standard deviation for each of the outcomes

 Discuss: Why would we use the mu and sigma notation here, instead of x bar and lowercase s? Do they represent statistics or parameters?

### 3.2 Mean Calculation Example

Car example from earlier:

| x | P(x) | x * P(x) |
|---|------|----------|
| 0 | 0.10 | 0.00     |
| 1 | 0.20 | 0.20     |
| 2 | 0.30 | 0.60     |
| 3 | 0.25 | 0.75     |
| 4 | 0.15 | 0.60     |

$$\mu = \sum (x * P(x)) = 2.15$$

### 3.3 Standard Deviation Calculation Example

Car example from earlier:

| x | P(x) | x * P(x) | (x - mu)^2 * P(x) |
|---|------|----------|-------------------|
| 0 | 0.10 | 0.00     | 0.462250          |
| 1 | 0.20 | 0.20     | 0.264500          |

| x | P(x) | x * P(x) | (x - mu)^2 * P(x) |
|---|------|----------|-------------------|
| 2 | 0.30 | 0.60     | 0.006750          |
| 3 | 0.25 | 0.75     | 0.180625          |
| 4 | 0.15 | 0.60     | 0.513375          |

$$\sigma = \sqrt{\sum (x - \mu)^2 * P(x)} = 1.1948$$

### 3.4 Section Conclusion

- Probability distribution functions serve a purpose - that is - they can describe a particular                      in the probability of outcomes in the data.
- Once we understand what pattern the variable fits, we can use this information for other                     , as there are the 4 specialty distributions mentioned earlier: geometric, hypergeometric, poisson, and binomial

## 4 Binomial Distribution

### 4.1 Introduction

- The binomial distribution has a certain,                      number of trials represented as  $n$ 
  - Each trial is                      and does not affect subsequent trials
- There are two possible                      in the binomial distribution, “success” and “failure”
  - $P(\text{success}) = p$
  - $P(\text{failure}) = q$
  - $p + q = 1$
- The experiment described above fits the **binomial probability distribution**. Where the random discrete variable  $X$  represents the number of successes obtain in  $n$  trials
- For the binomial probability distribution:

$$\mu = n * p$$

$$\sigma^2 = n * p * q$$

$$\sigma = \sqrt{n * p * q}$$

## 4.2 Notation

$$X \sim B(n, p)$$

- This formula is read as “X is a random variable with a binomial distribution”, where  $n$  and  $p$  represent the same things as before

## 5 Conclusion

### 5.1 Recap

- In this shorter lecture, we introduced the concept of discrete random variables, and how they can be represented with probability distribution functions
- We played around with calculating expected mean and standard deviations for outcomes from the probability distribution function table
- We introduced the binomial distribution as a special, specific pattern for a probability distribution represented via “successes” and “failures”

### 5.2 Lecture Check-in

- Make sure to complete and submit the lecture check-in